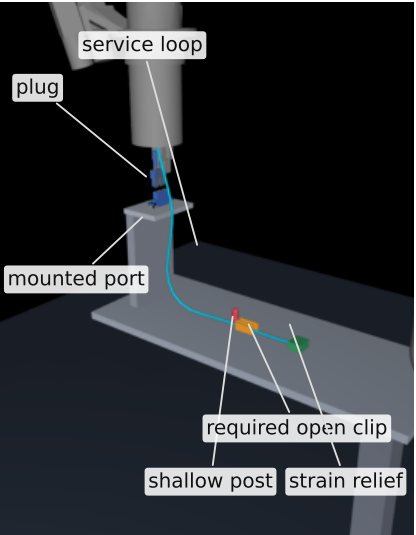
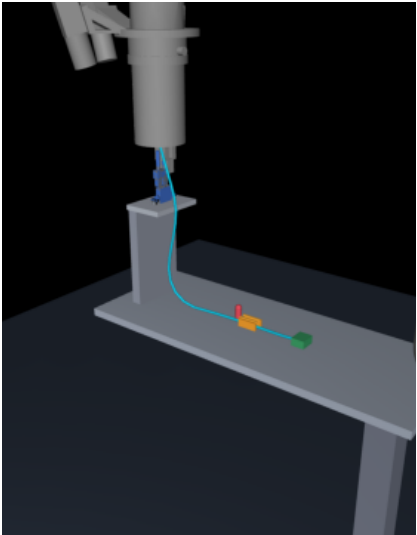


Constrained-cable connector recovery: held, clip-preserving seating (block v2 r01)

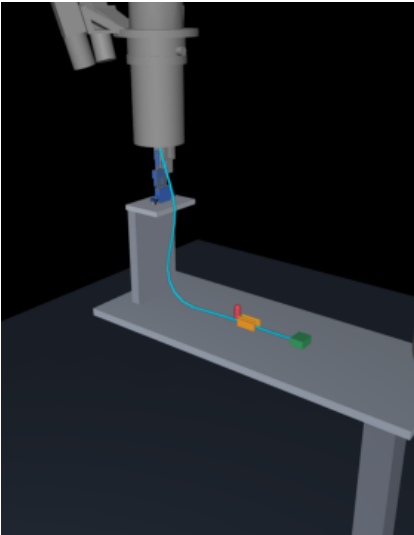
1 settled installation



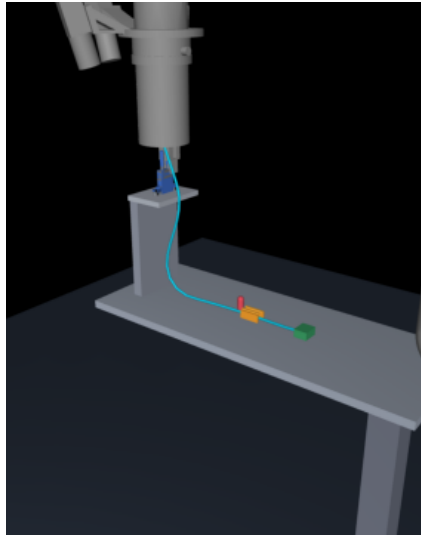
2 witnessed contact stall



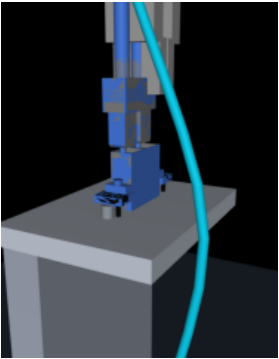
3 robot-driven repair begins



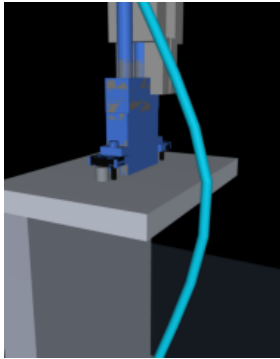
4 held clip-preserving seating



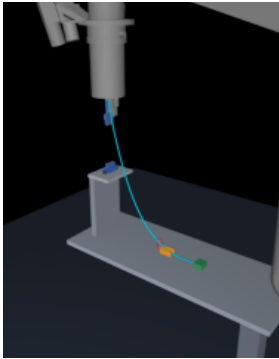
connector close-up at the witnessed stall



connector close-up at held seating



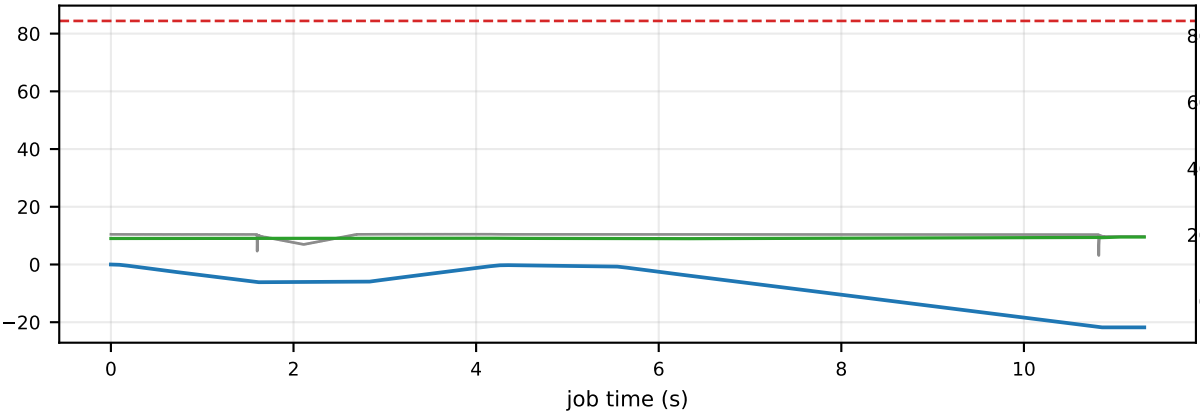
negative control: repair beyond the envelope



Simulation, MuJoCo 3.3.7, CPU. Frames rendered from recorded states.
Controller: blind existing insertion routine, then scripted cable-aware repair rules. No learned model of any kind is present.
Endpoint: robot still holds the plug, detector contact plus 0.5 s continuous seating dwell, required open clip retained throughout.
NOT a released or latched connection, learned pickup, or hardware result.

Measured clip-release travel: 84.4 mm of plug retreat.
Release is kinematic slack exhaustion; the strain-relief reaction stays below 0.2 N throughout. Playback: static frames, not real time.

blind attempt, witnessed stall, repair, held seating (g2_o5_rules)



same library, repair beyond the envelope (g1_bad_repair_near_home)

